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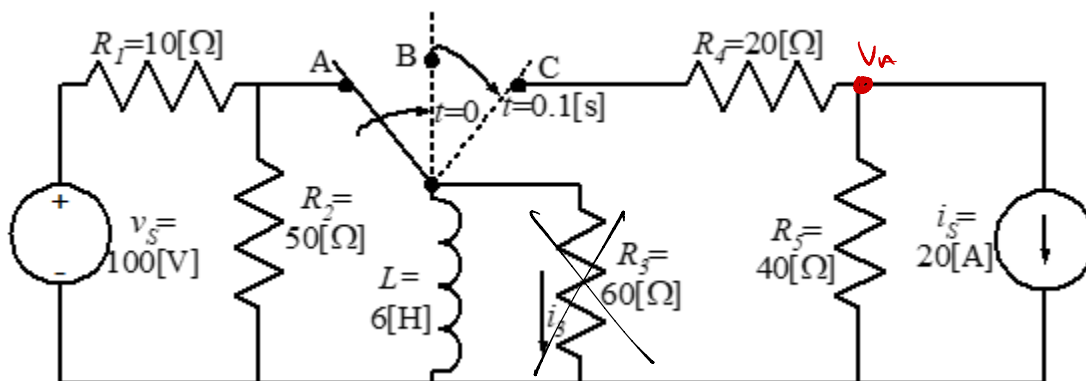
### ECE 2202 – HW 3

1) For the circuit shown below, the switch had been in position A for a long time.

At  $t = 0$  the switch moves to position B, and remains there for  $0.1[s]$ .

At  $t = 0.1[s]$  the switch moves to position C, and remains there.

For the time intervals  $0 < t < 0.1[s]$  and  $t > 0.1[s]$ , find the numerical equations for the current  $i_3(t)$ , where  $i_3$  is defined in the circuit shown.



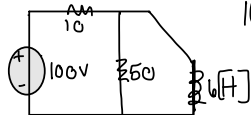
$$0 < t < 0.1[s]$$

$$Z = \frac{6}{10} = 0.6[s] \rightarrow 1.667[s]$$

$$i_3 = \frac{100[V]}{10[\Omega]} = 10[A]$$

$$i_L(\infty) = 0[A]$$

$$10[A]e^{-10(s)t}$$



$$0 < t < 0.1[s]$$

$$Z = \frac{L}{R} = \frac{6[H]}{50[\Omega]} = 0.12[s]$$

$$i_L(\infty) = 0[A]$$

$$i_L(0) = 10[A]$$

$$i_L(t) = i_L(\infty) + [i_L(0) - i_L(\infty)]e^{-\frac{t}{Z}}$$

$$0 < t < 0.1[s]$$

\* Find Req Not at steady state

$$i_3 = -i_L(t)$$

$$V_L = L \frac{di}{dt} = 6[H] \left( -13.33[A] + 17e^{-\frac{(t-0.1)}{2(s)}} \right)$$

$$\frac{d}{dt} \left( 102e^{-\frac{(t-0.1)}{2(s)}} \left( \frac{1}{0.2} \right) \right) = i_3$$

$$-510e^{\frac{1}{2}-5t} = i_3$$

$$-8.5[A]e^{-5(t-0.1)}$$

$$t > 0.1[s]$$

$$Z =$$

$$R_{eq} = (40 + 20) \parallel 60 = 30[\Omega]$$

$$\frac{L}{R_{eq}} = \frac{6[H]}{30[\Omega]} = 0.2[s]$$

$$i_L(0.1) = 3.67[A]$$

$$10e^{-10(s)t}$$

$$\frac{V_A}{20\Omega} + \frac{V_A}{40\Omega} + 20[A] = 0$$

$$i_L(\infty) = \frac{V_A}{20\Omega} = -13.33[A]$$

$$i_L(t) = -13.33 + [3.67 + 13.33]e^{-\frac{(t-0.1)}{0.2(s)}}$$

$$[A](-13.33 + 17.93)e^{-(t-0.1)/0.2(s)}$$

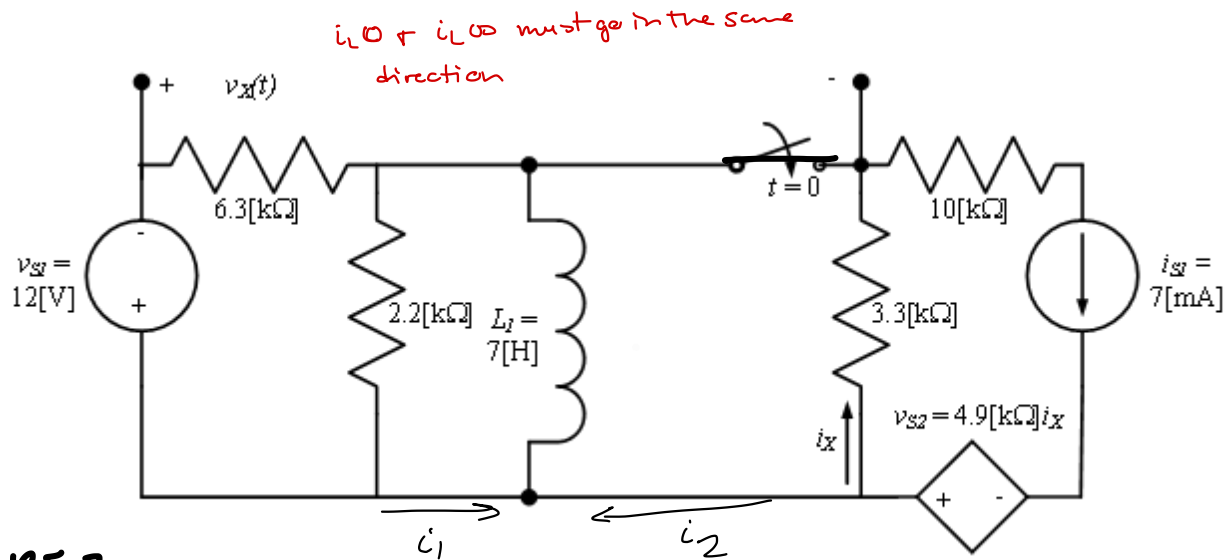
$$0 < t < 0.1[s]$$

$$(10e^{-10(s)t})[A]$$

$$t > 0.1[s]$$

$$-8.5[A]e^{-5(t-0.1)[s]}$$

2) Use the circuit below to solve. The switch was open for a long time before  $t = 0$ . Then, at  $t = 0$ , the switch closed. Find  $v_x(5[\text{ms}])$ .



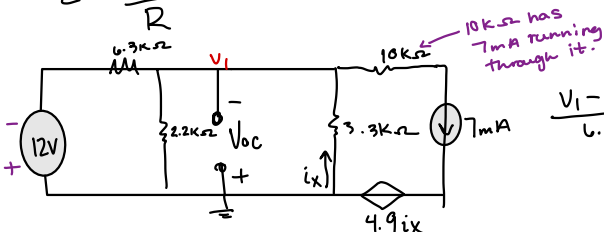
$$i_L(0^-) = \frac{12[\text{V}]}{6.3[\text{k}\Omega]} = 1.9[\text{mA}]$$

$KCL$

$$i_L(\infty) = i_{\infty} = i_1 + i_2$$

$$\frac{12[\text{V}]}{6.3[\text{k}\Omega]} + 7[\text{mA}] = 8.9[\text{mA}]$$

$$Z = \frac{L}{R}$$



$$\frac{V_1 - 12\text{V}}{6.3\text{k}\Omega} + \frac{V_1}{2.2\text{k}\Omega} + \frac{V_1}{3.3\text{k}\Omega} + 7\text{mA} = 0$$

$$V_1 = -9.718[\text{V}] \rightarrow V_{OC} = 9.718[\text{V}]$$

$$R_{Th} = \frac{V_{OC}}{I_{SC}} = \frac{9.718[\text{V}]}{?} = 1.09[\text{k}\Omega]$$

$$Z = \frac{7[\text{H}]}{1.09[\text{k}\Omega]} = 6.4[\text{ms}]$$

$$i_L(t) = 8.9[\text{mA}] + \frac{1.9[\text{mA}] - 8.9[\text{mA}]}{8.9 - 7e^{-\frac{t}{6.42[\text{ms}]}}} e^{-\frac{t}{6.42[\text{ms}]}}$$

$$v_x(5\text{ms}) = -12[\text{V}] + 3.5[\text{V}] = -8.5[\text{V}]$$

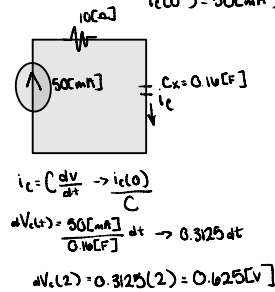
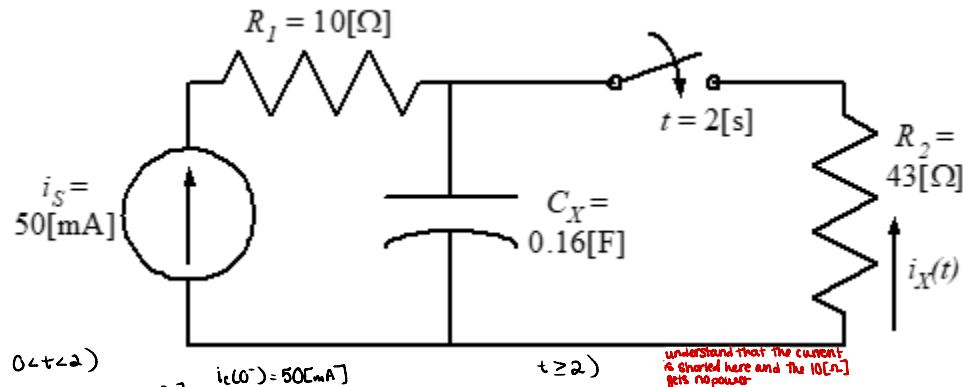
$$v_L(t) = C \frac{di_L}{dt} = 7 \cdot \frac{d}{dt} \left( 8.9 - 7e^{-\frac{t}{6.42[\text{ms}]}} \right)$$

$$= -\frac{7}{6.42[\text{ms}]} \cdot 7[\text{mA}] e^{-\frac{t}{6.42[\text{ms}]}} =$$

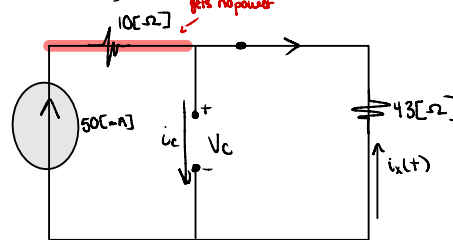
$$7632.4 e^{-\frac{t}{6.42[\text{ms}]}} \rightarrow$$

$$\approx 7.63 e^{-\frac{t}{6.42[\text{ms}]}} \rightarrow 7.63 e^{-\frac{5\text{ms}}{6.42[\text{ms}]}} = 3.5[\text{V}]$$

3) The energy stored in the capacitor in this circuit was zero at  $t = 0$ . Then, 2[s] later, the switch closed. Find  $i_X(4[s])$ .



$V_C(0) = 0[\text{V}]$   
 $V_C(2^-) = 0.05 \text{ V}$   
 this is before we get to 2[s]



$$i_X(t) = \frac{V_C(t)}{R_2} = \frac{V_C(t)}{43[\Omega]}$$

$$\tau = R_{eq} \cdot C = 43[\Omega] \cdot 0.16[\text{F}] \rightarrow 6.88 \text{ s}$$

$$V_C(t) = I_S R_2 + (V_C(0) - I_S R_2) e^{-\frac{t-2}{\tau}}$$

$$50[\text{mA}] \cdot 43[\Omega] = 2.15 \text{ V}$$

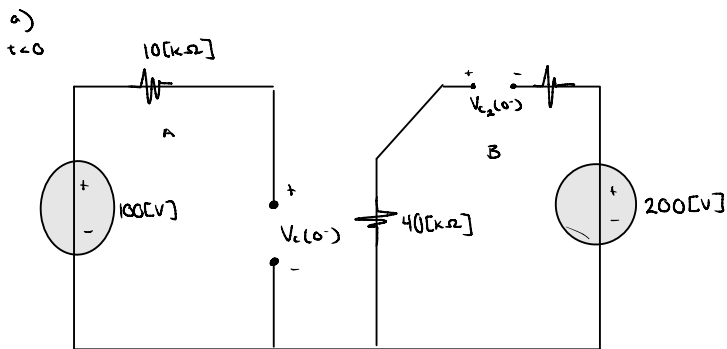
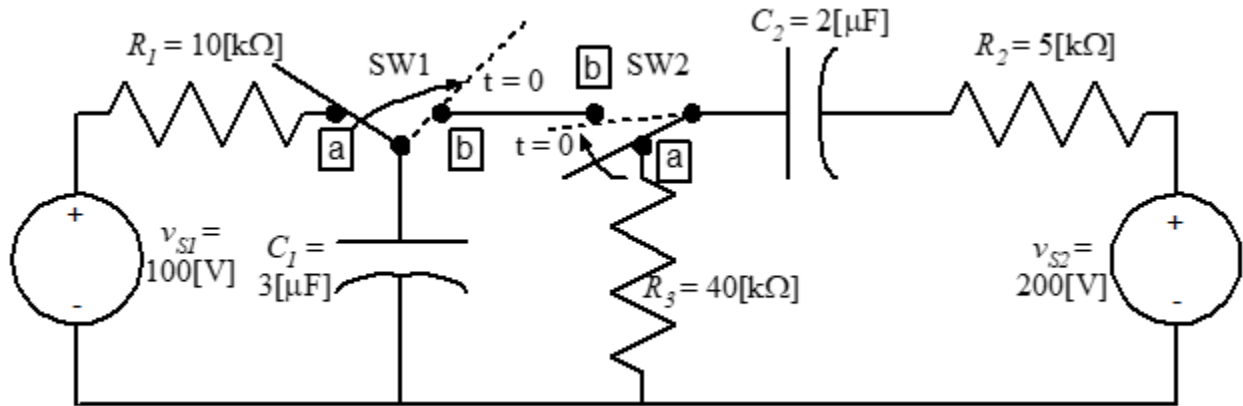
$$V_C(t) = 2.15 \text{ V} + (0.05 \text{ V} - 2.15 \text{ V}) e^{-\frac{t-2}{6.88}}$$

$$i_X(t) = \frac{V_C(t)}{R_2} = \frac{2.15 \text{ V} - 1.925 e^{-\frac{t-2}{6.88}}}{43[\Omega]}$$

$$i_X(4) = \frac{2.15 \text{ V} - 1.925 e^{-\frac{4-2}{6.88}}}{43[\Omega]} = -35.202 \text{ mA}$$

4) For the circuit shown below, switches SW1 and SW2 have been in position a for a long time. At  $t = 0$ , both switches are moved instantaneously and simultaneously to position b and remain there.

- Find the power delivered by the  $v_{S2}$  voltage source, as a function of time, for  $t > 0$ .
- Calculate the numerical value of the total energy stored in the capacitors at  $t = \infty$ .



b)

$$t = \infty$$

$$\text{Energy} = \frac{1}{2} CV^2$$

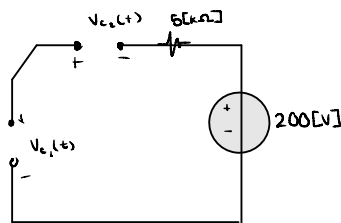
$$W = \frac{1}{2} (1.2 \mu\text{F}) (200\text{V})^2$$

$$= 0.024 \text{ [J]}$$

$V_{C1}(0^-) = 100\text{V}$        $V_{C2}(0^-) = 200\text{V}$

$$C_{eq} = \frac{C_1 C_2}{C_1 + C_2} = \frac{3 \mu\text{F} \cdot 2 \mu\text{F}}{3 \mu\text{F} + 2 \mu\text{F}} = 1.2 \mu\text{F}$$

$t \geq 0$



$T_c = 0.6\text{s}$        $V_{eq}(0) = 200 + 100 = 300\text{V}$

$$V_{eq}(0) = V_{C1} + V_{C2} = 300\text{V}$$

$$V_{C1}(t) = V_{C1}(\infty) + (V_{C1}(0) - V_{C1}(\infty)) e^{-\frac{t}{\tau}}$$

$\tau = RC_{eq}$

$$5\text{k}\Omega (1.2 \mu\text{F}) = 6\text{ms}$$

$$V_{eq}(t) = 200\text{V} + (300\text{V} - 200\text{V}) e^{-\frac{t}{6\text{ms}}}$$

$$V_{eq}(t) = 200\text{V} + (100\text{V} \cdot 4.14 e^{-73}) \text{ [V]}$$

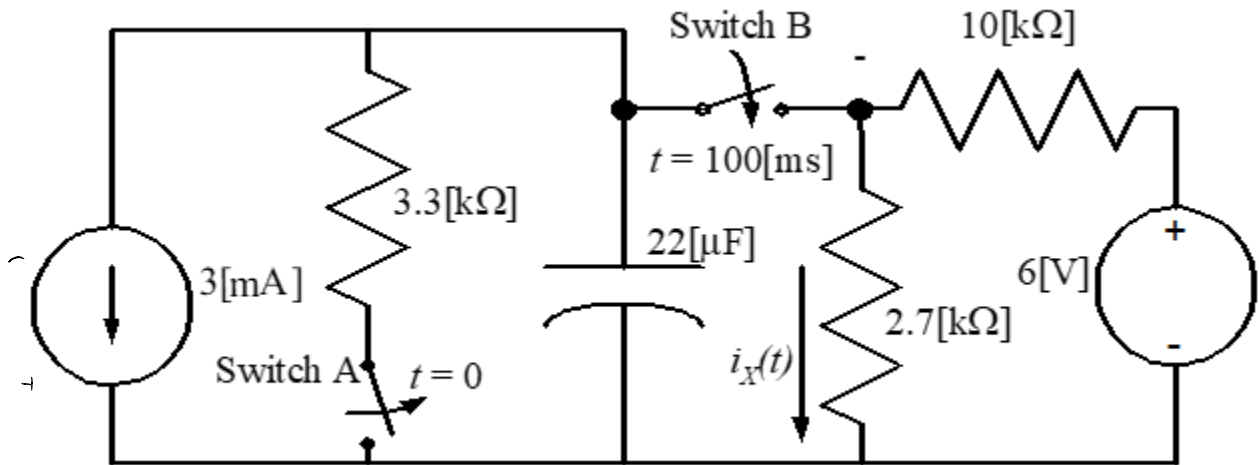
$$i_{Ceq} = C \frac{dV_{eq}(t)}{dt} = 1.2 \mu\text{F} \cdot \frac{d}{dt} (4.14 e^{-\frac{t}{6\text{ms}}})$$

$$\approx -0.02 e^{-\frac{t}{6\text{ms}}} \text{ [A]}$$

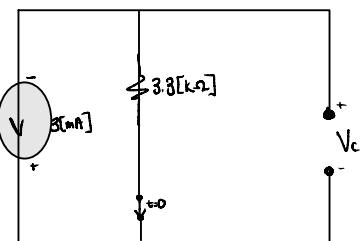
$P_{del \text{ by } V_{C2}(t)} = 200\text{V} \cdot i_{Ceq}(t)$

$$= -4 e^{-\frac{t}{6\text{ms}}}$$

5) For the circuit shown below, Switch A had been closed for a long time, and Switch B had been open for a long time, before  $t = 0$ . At  $t = 0$ , Switch A opened. Then, 100[ms] later, Switch B closed. Find  $i_X(200[\text{ms}])$ .



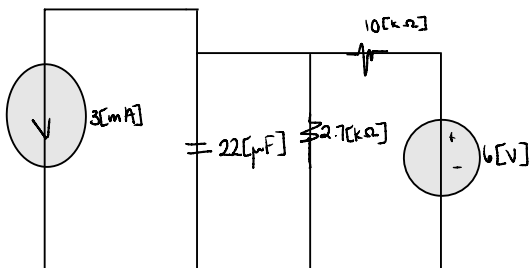
$t < 0$



$$V_C(0^-) = 3.3[\text{k}\Omega] \cdot (-3[\text{mA}])$$

$$V_C(0^-) = -9.9[\text{V}]$$

$0 < t < 100[\text{ms}]$



$$V_C(t) = \frac{1}{C} \int_0^t i_C(t) dt + V_C(t_0)$$

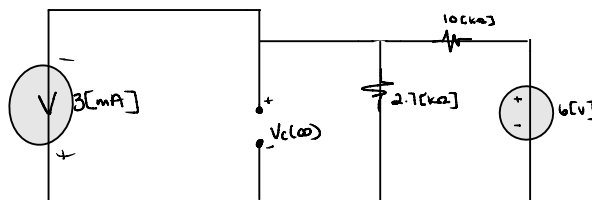
$$C = 22[\mu\text{F}]$$

$$i_C(0) = -3[\text{mA}]$$

$$V_C(0) = -9.9[\text{V}]$$

$$\frac{1}{22[\mu\text{F}]} \int_0^{0.1[\text{s}]} -3[\text{mA}] dt - 9.9[\text{V}] = V_C(0.1[\text{s}]) = -23.5[\text{V}]$$

$t \geq 100[\text{ms}]$



$$3[\text{mA}] + \frac{V_C(\infty)}{2.7[\text{k}\Omega]} + \frac{V_C(\infty) - 6[\text{V}]}{10[\text{k}\Omega]} = 0$$

$$V_C(\infty) = -5.10[\text{V}]$$

$$Z = R_{eq}C$$

↓

$$10[\text{k}\Omega] \parallel 2.7[\text{k}\Omega] = \frac{(2.7 \cdot 10)[\text{k}\Omega]}{(2.7 + 10)[\text{k}\Omega]} = 2125.98[\Omega] (22[\mu\text{F}])$$

$$= 0.0467[\text{s}]$$

$$T_0 = 100[\text{ms}]$$

$$V_C(0.1[\text{s}]) = -23.5[\text{V}]$$

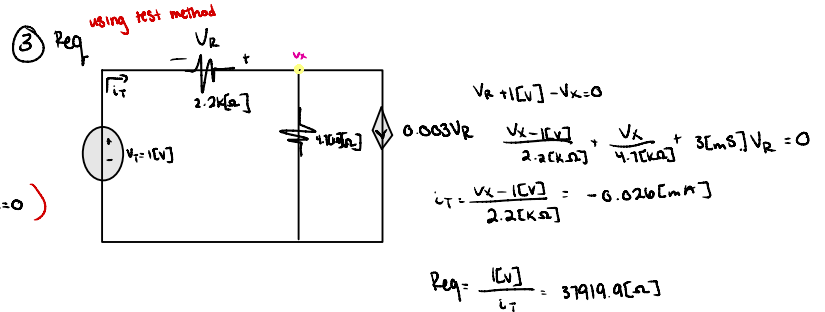
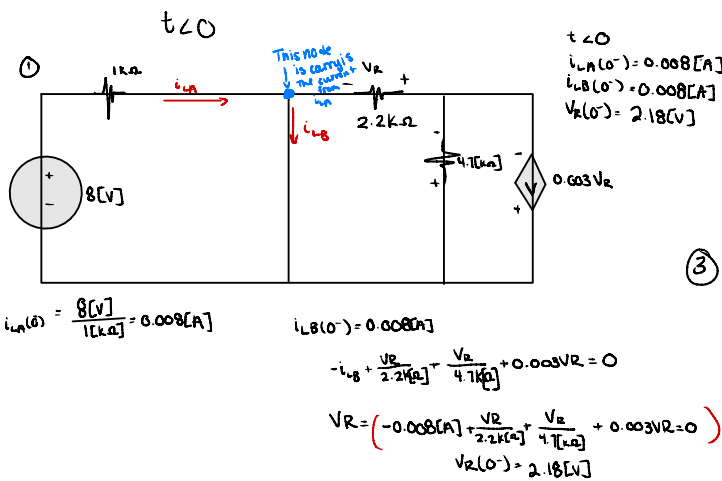
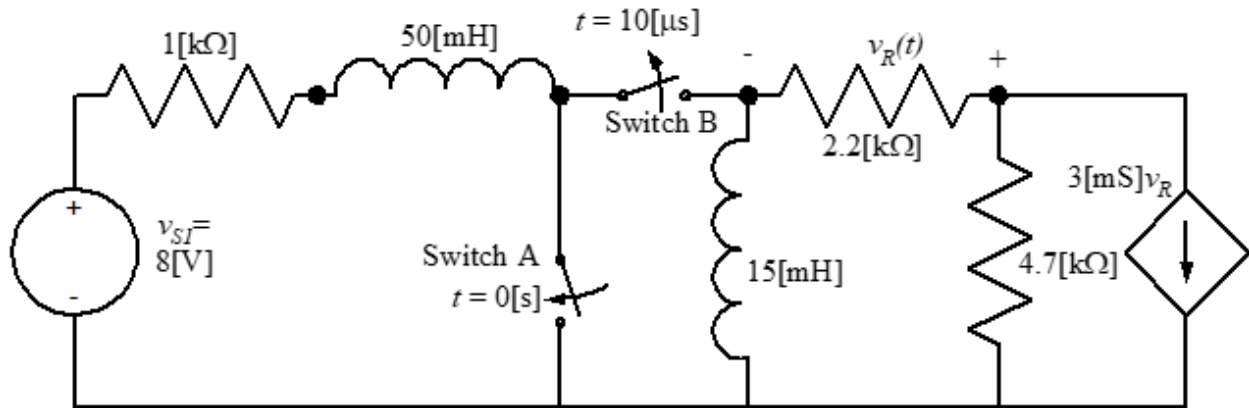
$$V_C(\infty) = -5.10[\text{V}]$$

$$V_C(t) = -5.1[\text{V}] - 18.4 e^{\frac{t - 0.1}{0.0467}}$$

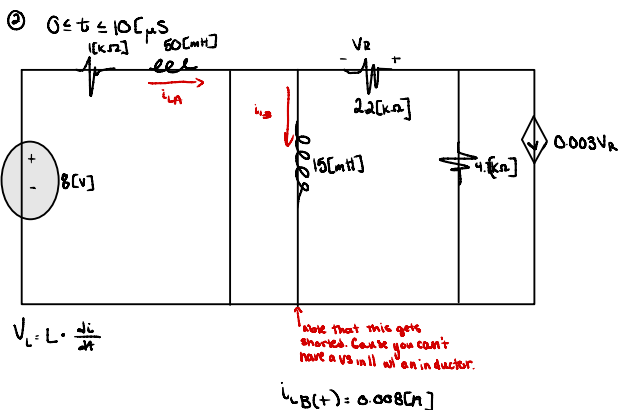
$$i_X(200[\text{ms}]) = \frac{(-5.1[\text{V}] - 18.4 e^{\frac{-0.2 - 0.1}{0.0467}})}{2.7[\text{k}\Omega]} [\text{V}]$$

$$i_X(200[\text{ms}]) = -2.70[\text{mA}]$$

6) For the circuit shown below, the Switch A was open for a long time, and Switch B was closed for a long time, before  $t = 0$ . At  $t = 0$ , Switch A was closed. Then,  $10[\mu\text{s}]$  later, Switch B was opened. Find  $v_R(11[\mu\text{s}])$ .



④  $Z = \frac{s}{R} \rightarrow \frac{15[\text{mH}]}{37919.9[\Omega]} = 3.955[\text{E}-7][\text{s}]$



⑤  $V_R(t) = i_{L8}(t) \cdot 2.2[\text{k}\Omega] = 17.16[\text{V}]$

$V_R(t) = i_{L8}(11[\mu\text{s}]) \cdot 2.2[\text{k}\Omega]$

$i_{L8} = i_{L8}(t_0) e^{-\frac{(t-t_0)}{\tau}}$

$0.008[\text{A}] e^{-\frac{-(11\mu\text{s} - 10\mu\text{s})}{3.955[\text{E}-7][\text{s}]}} = 8.699[\text{mA}] \cdot 2.2[\text{k}\Omega]$

$V_R(11\mu\text{s}) = 1.905[\text{V}]$

$i_{L8}(t) = i_{L8}(\infty) + (i_{L8}(t_0) - i_{L8}(\infty)) e^{-\frac{(t-t_0)}{\tau}}$

$t_0 = 10[\mu\text{s}] \rightarrow 10\text{E}-6[\text{s}]$

$i_{L8}(t_0) = i_{L8}(10\mu\text{s}) = 0.008[\text{A}]$

$\tau = \frac{1}{R} \rightarrow$

## Solutions

1.

**For  $0 < t < 0.1$  [s]:**

$$i_3(t) = -10e^{-10t}[\text{A}] \quad i_3 t = -10e^{-10t}[\text{A}]$$

**For  $t > 0.1$  [s] :**

$$i_3(t) = -8.5e^{-5(t-0.1)}[\text{A}] \quad i_3 t = -8.5e^{-5(t-0.1)}[\text{A}]$$

2. **-8.50[V]**

3. **-23.5[mA]**

4. a)  **$p_{del,vs2}(t) = -4e^{-166.67t}[\text{W}]$**  b) **25[mJ]**

5. **-2.70[mA]**

6. **1.405[V]**